2.3 Driver Information in Work Zones

Processing roadway information in a timely manner is essential to driving safely through a work zone. Inappropriate or unclear messages and/or the incorrect placement of signs, markings, and other devices may mislead and confuse drivers.

There are usually three types of **traffic control messages** in work zones:

1. Warnings about potential hazards
2. Postings of maximum speeds and applicable regulations
3. Delineation of the traveled roadway

Positive guidance for drivers is the process of giving **clear, correct, accurate, relevant, and timely information, and visual cues** through the work zone by means of temporary traffic control devices.

2.3.1 Manage Driver Expectations

There are three types of driver expectations:

1. **Continuity Expectation.** Drivers expect that traffic control devices used in one situation will be used in all situations. For example, a Traffic Control Person Ahead sign is always used in advance of a Traffic Control Person.
2. **Event Expectation.** Drivers expect that if something has not happened, it will not happen. Over time, drivers develop an expectation of various unique events that occur on roadways.
3. **Timing Expectation.** For events that recur in cycles, drivers expect that the longer a state occurs, the more likely it is to change. For example, a stale green signal display can be expected to turn yellow and then red.

Providing consistent driver information will evoke consistent driver behaviour. Displaying the same accurate message repeatedly motivates drivers to take the same action each time they see the sign or device.

2.3.2 Manage Driver Workload

Driver work load relates to the ability for drivers to process information. Driver workload increases through work zones. Providing too much information can result in one of three driver responses:

1. **Doing the wrong thing.** A driver turns in the wrong direction because the more important directional sign is lost among various other signs.
2. **Slowing or stopping.** A driver is distracted by the work activity or overwhelmed with too many signs and slows or stops to try and process the information.
3. **Taking a chance.** A driver is confused but continues and may or may not make the correct decision.

2.3.3 Manage Driver Sight Distances

Driver sight distance is the distance that a driver can see unobstructed along the highway. Driver sight distances are required for the driver to see and respond to conditions through the work zone. The required sight distance increases as the speed of the vehicle increases.

Positioning of devices should always consider the available sightlines of an approaching vehicle so road users can adequately respond.

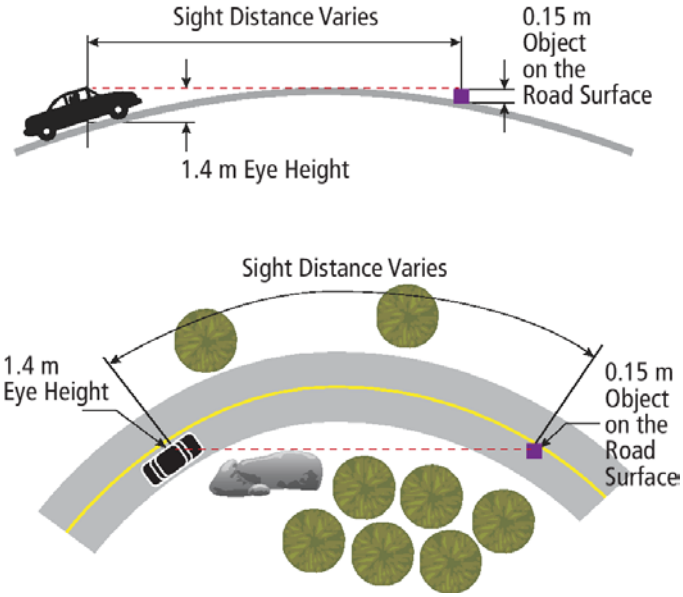


Figure 2.5: Driver Sight Lines

2.4 Management of Speed

Not all road work requires a reduction in the posted speed. Temporary traffic control devices are generally more effective than construction speed limits in drawing attention to the need for caution through work zones. A decision to reduce the existing posted speed limit should be clearly justified.

2.4.1 Basic Guidelines

Speed zones are effective when they appear reasonable to drivers. To make reduced speed zones credible and acceptable:

- Maintain consistency by complying with the standards in this Manual.
- Speed zones remain short and relevant to the nature of the work.
- Ensure that all speed zone signs are covered or removed when not required.

Reduced speed zones should be used:

- In conjunction with other signs or devices required for site-specific conditions
- To complement more effective traffic control
- When temporary road conditions are unsafe at the existing posted speed

To maintain driver cooperation, avoid unnecessarily restrictive requirements such as excessive length, excessive hours for which the zone is in effect, and specified speeds which are too low. The length of the speed zone is typically less than the length of the work zone.

When a further speed reduction is required within a reduced speed zone, signs and devices should be installed based on the higher speed. See *Section 6.6: Positioning of Temporary Traffic Control Devices* for details.

2.4.2 Construction Speed Zones

A Construction Speed Zone is established through a Traffic Management Plan to reflect hazards due to the nature and the type of work taking place.

A Construction Speed Zone requires the authorization of the Road Authority and is established by regulatory signs. It may be in place 24 hours a day, so it is important to ensure that it is appropriate and effective during inactive work. The Construction Speed Zone shall be modified or removed as needed.



R-003 C-080-T | R-004 C-080-T | C-082 | C-088 | C-086-1 OR C-086-2

Figure 2.6: Typical Construction Speed Zone Start and End Signage

1. Construction Speed Zones are identified with Maximum Speed R-004 signs and Construction Speed Zone C-080-T.
2. The Min $196 Fine – Speeding in Work Zones C-082 sign is a speed management tool for use when drivers fail to adhere to the speed limit.
3. The Work Zone Ends C-088 sign marks the end of the work zone.
4. The Thank You Resume Speed C-086-1 or C-086-2 sign may follow C-088 sign to advise drivers that they may resume the regulated posted speed. A Maximum Speed Limit R-004 sign showing the regular posted speed may be installed.

2.4.3 Temporary Speed Zones

A Temporary Speed Zone is a short duration speed reduction typically used for maintenance, surveying, and other work which does not have a significant impact to the roadway yet workers are present on, or adjacent to, the road surface.

Temporary Speed Zones require the approval of an onsite Supervisor, and are signed by a Survey Crew Working C-002-1 or a Crew Working C-002-2 sign, and show the appropriate speed for the work.

The C-002-2 may be used as a supplement to an R-004 Construction Speed Limit to remind drivers of the reduced speed as they approach the work area.

The end of a Temporary Speed Zone is marked with either a Thank You Resume Speed C-086-1 sign or a Maximum Speed R-004 sign. The temporary speed zone shall be removed when no longer required.



C-002-1 C-002-2 C-086-1

Figure 2.7: Typical Temporary Speed Zone Start and End Signage

2.4.4 Determining the Need for a Speed Reduction

Effective work zone speeds depend on the type of work, the change in road characteristics, and the proximity of workers to adjacent traffic. The construction speed limit shall not exceed the regular posted speed limit.

Obtain a clear understanding of these factors to determine appropriate speed zones:

- type of work and traffic control (traffic control persons or signals)
- roadway classification and surface
- type of traffic and volumes
- accesses and land use
- sight distances (fog, smoke, or dust)
- detours

On roadways with a regular posted speed limit of 80 to 120 km/h, consider reducing the speed limit when:

- Workers on foot are less than 6 metres from traffic with no intervening physical barrier, and average daily traffic volumes exceed 12,000 vehicles per day.
- There are changed traffic conditions on the site, such as detours, a reduced number of lanes, and varying surfaces.
- There are work vehicles entering or leaving the roadway.
- TCPs are directing traffic for planned work, the construction speed limit shall be ≤ 70 km/h.

On roadways with a regular posted speed limit of 50 to 70 km/h, consider reducing the speed limit when:

- Workers on foot are less than 3 metres from traffic with no physical barriers.
- There is significant interaction between work vehicles and through traffic.
- There is a reduced standard of alignment due to the road works.
- There is a loose surface such as gravel or a newly sprayed bitumen seal.
- Traffic is adjacent to an excavation.

On roadways with a regular posted speed limit below 50 km/h, consider reducing the speed limit when:

- There is a severe change in alignment.
- The work area is in a high-volume urban area where a variety of activities are competing for drivers' attention.

2.4.5 Transition Speed Zones

There are two options for signing transition speed zones:

Option 1 – Stepped Speed Zone Drop

Transition speed zones, approximately 500 m long are used when the reduction to the speed limit is greater than 30 km/h. For example:

- For a drop of 40 km/h, the transition speed should be 20 km/h above the Construction Speed Zone limit.
- For a drop of 60 km/h, the transition speed should be 30 km/h above the Construction Speed Zone limit.

Option 2 – Singular Speed Zone Drop using Additional Warning

An alternative approach to signing construction speed limits with a speed reduction > 30 km/h and ≤ 50 km/h is to provide additional advance warning through the installation of a Construction Speed Limit Ahead C-128 sign and a Distance C-130 T tab.

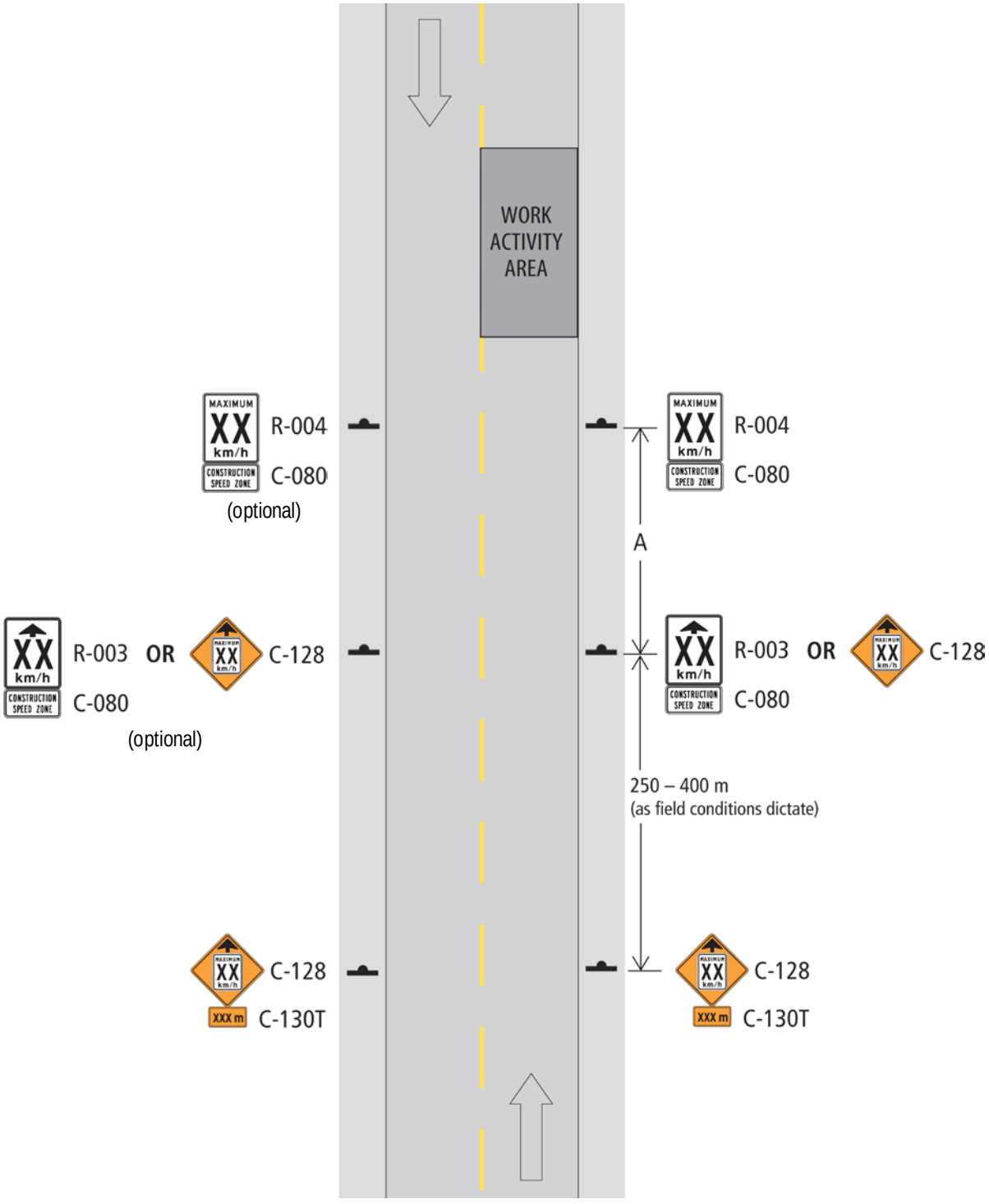
These signs are installed in advance of the Speed Limit Ahead R-003 (or additional C-128 sign) and Speed Limit R-004 signs establishing the construction speed limit. *Figure 2.8: Singular Speed Zone Drop Using Additional Warning* shows an example of how these signs should be used.

This process provides consistent messaging of the construction speed limit and replaces the transition speed zone. It suits work zones which have a distinct change in the roadway environment wherein a transition speed zone may not appear reasonable to a driver. See Technical Circular T-09/14 Speed Zone Transition Signs for more information.



C-128
C-130-T

Figure 2.8: Singular Speed Zone Drop Using Additional Warning



2.4.6 Assessing Speed Zone Effectiveness

If drivers are disregarding the construction speed zone, consider re-evaluating the following items.

- **Site characteristics:** Speed zone signs are clearly visible amongst other temporary traffic control devices. The grade of the roadway may necessitate additional advance warning.
- **Placement of speed zone signs:** The construction speed limit signs should be positioned approximately 10 seconds of travel time in advance of the work activity area.

Table 2.1: Distance Travelled Based on Speed and Time

Construction Speed (km/h)	10 Seconds of Travel Time
50	140 metres
60	170 metres
70	200 metres
80	220 metres
90	250 metres
100	280 metres
110	310 metres
120	330 metres

- **Unnecessarily long speed zones:** The construction speed zone should only encompass the area where the hazards exist from the work. Speed zones should be kept as short as practicably possible.
- **Speed zone speed limit too fast:** Drivers approaching the work zone too quickly and react unsafely. This could be a sign of inadequate perception time, stopping distance, and interpretation of the traffic control.
- **Speed zone speed limit too slow:** Driver disrespect of the construction speed limit leads to heavy queuing, improper passing or tailgating. The construction speed limit should always reflect the actions of a reasonable driver. A reasonable speed limit builds respect for all traffic control, improving overall safety through the work zone.
- **Conspicuousness of speed zone signs**: Keep traffic control simple, positive, and directive, with little need for drivers to make choices, minimizing driver workload. Speed zones signs are regulatory and therefore of high importance when placing traffic control devices.
- **Relevance of speed zone:** Reduced speed zone signs must be removed, turned, or covered when they are not needed.

2.4.7 "Slow Down, Move Over" Regulation and Work Zones

The "Slow Down, Move Over" Regulation in the Motor Vehicle Act requires that drivers reduce their speed when approaching or passing an official vehicle. In addition, the driver must move their vehicle into another lane if safe to do so.

An official vehicle is one which displays a flashing red, blue, white, or amber light.

The Motor Vehicle Act also requires that when work is occurring on a roadway, traffic control devices shall be installed to indicate that persons or equipment are on the roadway. Therefore, for planned and scheduled construction/maintenance activities, the Slow Down, Move Over Regulation does not replace the necessity to establish temporary traffic control, including a construction speed limit, if warranted.

2.4.8 Highway Work Zones – Split Direction of Travel

On divided highways or multilane roadways where the work zone is impacting one direction of travel, the Road Authority may establish a split speed zone where the speed limit differs for each direction of travel.

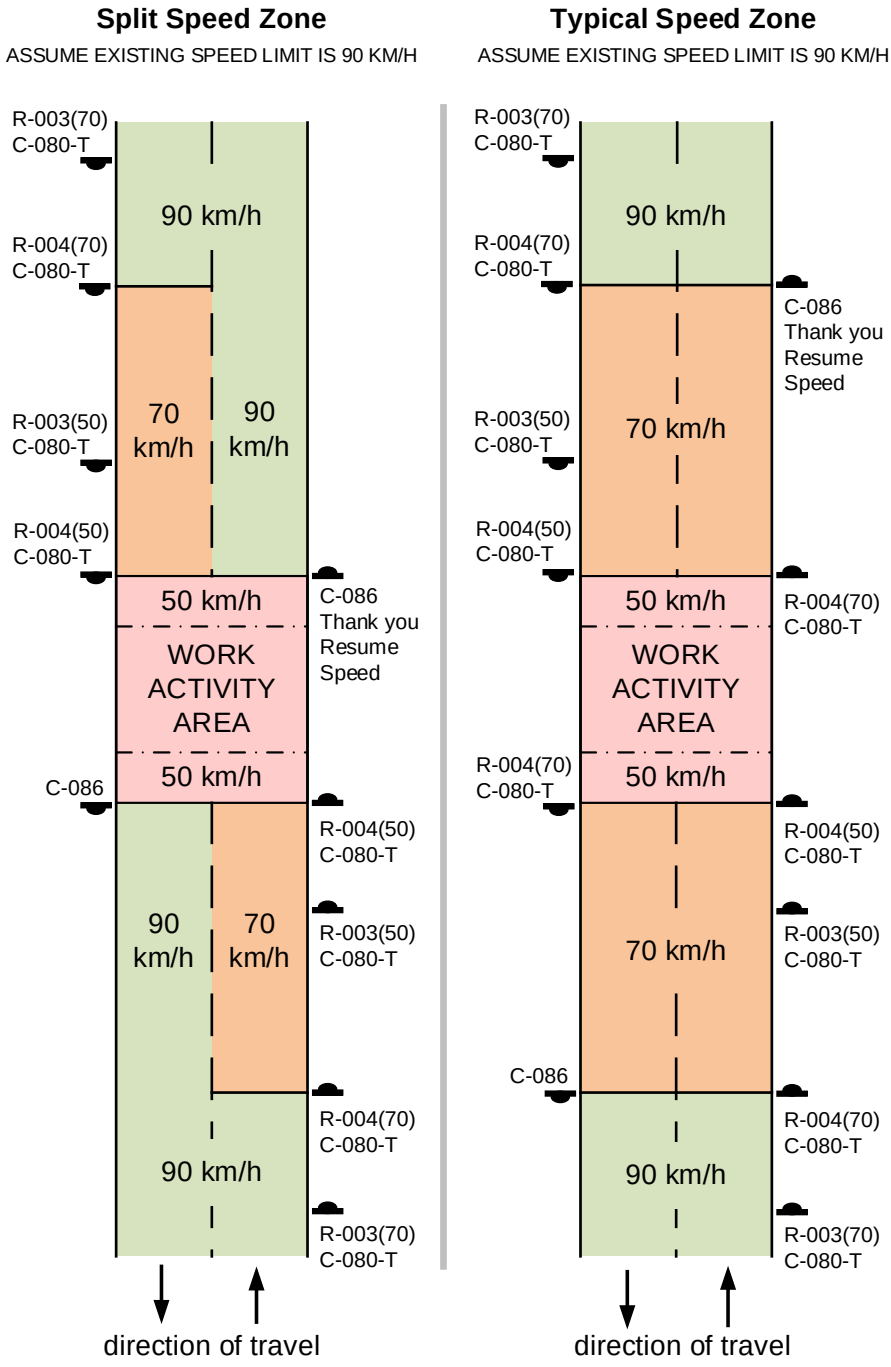


Figure 2.9: Split Speed Zone and Typical Speed Zone

2.5 Establishing Work on Roadways

This section outlines the requirements and processes used by the Road Authority and Prime Contractor in establishing and approving works on roadways.

2.5.1 Road Authority Requirements for Work

The Road Authority sets out the requirements of the Prime Contractor to be accommodated in the Traffic Management Plan.

Each Road Authority has its own policies, specifications, requirements, and systems for Traffic Management Plans. These may include:

- hours of work, acceptable delay, and lane closures
- acceptable temporary traffic control devices
- notification requirements
- municipal approval
- review requirements
- requirements for plan review, updates and modifications
- minimum requirements for Traffic Management Plan documentation

For Ministry projects, these requirements are developed by District Staff (Area Manager, Operations Manager, etc.) and, in more complex situations, in consultation with Traffic Engineering. The District Manager of Transportation, or designate, shall approve the closure of travel lanes for any road works on Provincial Highways.

For non-Ministry initiated projects, it is the responsibility of the agency, group, or other Road Authority engaging the work to define the Strategy and provide it to the Ministry for authorization to close the road and/or commence the work (see *Appendix E: Lane Closure Request Form).*

2.5.2 Traffic Management Plan Development

The Prime Contractor shall develop the Traffic Management Plan in accordance with *Section 3: Traffic Management Plans*, which provides detailed requirements for Category 1, 2, and 3 Traffic Management Plans.

The Traffic Management Plan shall include all requirements set out by the Road Authority.

2.5.3 Road Authority Acceptance

No work may be performed on the roadway unless the Prime Contractor's Traffic Management Plan meets the Road Authority's requirements and has been reviewed and accepted by the Road Authority.

The process of documenting and accepting the Traffic Management Plan generally follows:

- The Prime Contractor designates a qualified Traffic Control Manager or Engineer for the project and that person designs and approves the Traffic Management Plan for the Prime Contractor.
- The Road Authority reviews the Prime Contractor's Traffic Management Plan. If it is acceptable, authorization is granted to proceed with the work. Authorization does not constitute formal approval of the details in the Traffic Management Plan. The Traffic Management Plan is reviewed for completeness but the Prime Contractor is responsible for the content.
- If the Road Authority has concerns about the Traffic Management Plan, it may be returned to the Prime Contractor for amendment and re-submission.

For Ministry projects, all plans requiring Ministry acceptance shall be submitted to the Ministry Representative, who provides them to the District Manager, Transportation or designate. Before submitting a Traffic Management Plan to the Ministry, the Prime Contractor shall receive stakeholder approval, including municipalities and/or regional districts, for the impact of the work.

For non-Ministry projects that require access to a Ministry right-of-way, notification shall be provided directly to the District Manager, Transportation or designate.

2.5.4 Compliance

The Prime Contractor shall perform all works in compliance with the Ministry-accepted Traffic Management Plan for the project.

The Traffic Control Plan component of the Traffic Management Plan shall comply with the requirements of Part 18 of WorkSafeBC's Occupational Health and Safety Regulation.

2.5.5 Traffic Management Plan Documentation

Traffic management documentation is the responsibility of the Prime Contractor. The documentation process includes, but is not limited to:

- amending the Traffic Management Plan as the site requires
- keeping a record of all temporary traffic control devices used on the project
- recording the daily status of temporary traffic control devices and times of device changes
- recording the status of temporary traffic control devices as soon as practicable after any incident (including motor vehicle), taking appropriate measurements and photographs, and marking them with the date and time

2.5.6 Traffic Management Plan Resubmission

The Prime Contractor shall amend and re-submit the Traffic Management Plan to the Ministry for review and acceptance whenever:

- a change to work activities alters traffic management requirements
- a change to the scheduling of work activities is proposed
- a change to the traffic management plan is proposed

If a Traffic Management Plan is signed and sealed by a Traffic Engineer, any changes that significantly impact the planned Traffic Operations should be relayed to the Traffic Engineer

2.6 Quality Management

2.6.1 Quality Control

Quality Control (QC) is the responsibility of the **Prime Contractor**. It is documented in the Prime Contractor's Quality Control Plan.

It is an inspection process that examines the quality and effectiveness of the temporary traffic control.

2.6.2 Quality Assurance

Quality Assurance (QA) is the responsibility of the **Road Authority**.

QA is a process which ensures the Prime Contractor is following their Quality Control Plan.

It includes reviewing the Prime Contractor's Traffic Management Plan prior to the commencement of road works and conducting periodic reviews of traffic control as work progresses.

The Road Authority's QA process includes:

- **verifying** Traffic Management Plans to ensure they meet regulatory, contractual, and project-specific requirements
- **reviewing** Traffic Control Plans for overall effectiveness through the work zone.
- **recording** deficiencies and presenting them to the Prime Contractor for review and correction

2.6.3 Traffic Management Audits

Traffic Management Plan and Traffic Control Plan audits are typically conducted by the Road Authority (or those under contract to do so) when reviewing the documentation and field layout of the Prime Contractor. They may also be used by Road Authorities or Prime Contractors to assist in developing Traffic Management Plans.

Section 3.7 Auditing Traffic Management Plans details the two audit forms found in *Appendix D*:

- *Traffic Management Plan Documentation Audit Form*
- *Traffic Management Plan Field Audit Form*

PART A – TRAFFIC MANAGEMENT

Section 3: Traffic Management Plans

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PART A – TRAFFIC MANAGEMENT

Section 3: Traffic Management Plans

3.1 Introduction

Traffic Management Plan	The Prime Contractor's project-specific plan that details the strategies for protecting workers and safely and efficiently moving road users through the work zone, including any requirements of the Road Authority.

The Traffic Management Plan is developed by first assigning a Project Category (Category 1, 2, or 3). The Project Category determines the extent of information required in the Traffic Management Plan and sub-plans. The Traffic Management Plan is then written based on these requirements.

The Traffic Management Plan includes up to four sub-plans:

- Traffic Control Plan[1]
- Incident Management Plan
- Public Information Plan
- Implementation Plan

The Traffic Management Plan combines these sub-plans into a single document that demonstrates a full understanding of the site-specific issues and project requirements.

It should also contain provisions for updating the Traffic Management Plan and sub-plans throughout the project to address issues as they arise.

Additional documents may be referred to when developing a Traffic Management Plan:

- *Appendix C: Templates for Traffic Management Plans*
- *Appendix D: Traffic Management Plan Audit Forms*

[1] A Traffic Control Plan is required for all planned work on Provincial roadways.

3.2 Traffic Management Sub-Plans

3.2.1 Traffic Control Plan

The Traffic Control plan outlines the temporary traffic control devices used to protect workers and move road users safely through the work zone. A Traffic Control Plan is required for all planned work on or near Provincial roadways.

Traffic Control Plans range in scope from being very detailed to simply using typical drawings contained in this Manual. The degree of detail in the Traffic Control Plan depends entirely on the complexity of the work and road environment.

For longer-term projects, multiple traffic control plans may be required as the work progresses. These plans shall be approved by the Prime Contractor and presented for acceptance by the Road Authority.

A Traffic Control Plan outlines the specific traffic control devices that will be used on the project, how they will be implemented, and on what schedule. It does this by using a combination of:

- **text descriptions** of the location of the work zone, proposed work activities, proposed traffic control measures, and the specific times and dates when work will be undertaken.
- **customized traffic control layouts** are sketches of the traffic control layouts found in this manual and modified to suit site specific requirements. .
- **customized drawings** (scale drawings) show all existing roadway geometry and features as well as all temporary traffic control devices and layouts.

3.2.2 Incident Management Plan

An Incident Management Plan identifies the Prime Contractor's procedures for responding to unplanned events or incidents. An incident includes events which affect Traffic Operations for workers and/or the travelling public, such as a collision which occurs within the construction zone or unexpected adverse weather conditions.

The Incident Management Plan should define a process of regular review and analysis for identifying actions that will reduce the frequency and severity of incidents.

It is important to respond appropriately to an individual incident, identify actions for reducing the occurrence of incidents, and minimize their impact on workers and traffic.

The **Incident Management Plan** includes procedures for:

- identifying and verifying that an incident has occurred
- quickly responding to an incident to ensure worker and public safety
- contacting appropriate emergency response agencies and Road Authority personnel
- ensuring that emergency responders are assisted by project staff
- quickly assessing an incident's impact on traffic operations and identifying the appropriate response measures

- routing traffic away from the incident by means such as using a diversion or detour around the work site
- ensuring that the Road Authority is updated regarding worker and public safety, traffic conditions, and actions taken to normalize traffic flow
- ensuring that travellers within the traffic queue are made aware of the event(s), their options, and estimated time of opening
- ensuring that any health or safety issues from workers or travellers in the traffic queue that may arise can be attended to
- taking action to restore normal traffic operations by modifying work plans and activities where necessary
- reviewing and analyzing incidents if they occur to identify causes and preventative actions, if any
- recording incident details and reporting them to the Road Authority

It shall also address the potential requirement for **emergency vehicles** to pass through the work zone, and processes for ensuring that:

- Emergency services will be contacted and advised of the planned work, potential issues regarding passage through the work zone, and viable alternative routes.
- Work operations will be stopped and the work zone cleared to allow emergency responders to pass queued traffic and move unimpeded through the work zone.
- Emergency responders in personal vehicles will be able to pass through the work zone safely and expediently.

3.2.3 Public Information Plan

A Prime Contractor's Public Information Plan identifies actions and procedures for informing the travelling public, project stakeholders, and the Road Authority of current traffic operations and planned changes. The significance of the work should dictate the degree of public communication and advance notice required.

The Public Information Plan should ensure that:

- communication issues are identified
- stakeholders are aware of the project and its impacts
- groups directly impacted by specific projects (local residents, businesses, etc.) are appropriately informed
- emergency response agencies and the travelling public are informed in a timely manner regarding work

3.2.4 Implementation Plan

An Implementation Plan identifies responsibilities and procedures for ensuring the other Traffic Management sub-plans are developed and implemented in a coordinated manner.

It also identifies the qualifications, responsibilities, and duties of the supervisory and management personnel who will implement the Traffic Management Plan, including the Traffic Control Manager and the Traffic Control Supervisor.

3.3 Project Category Determination

A structured process is used to determine the Project Category.

1. **Initial Category Assessment** — Assess the roadway and traffic features.
2. **Risk Analysis** — Identify the project-specific risks.
3. **Final Category Determination** — Combine the initial project assessment with the risk analysis to determine the final project category.

Project Categories are defined as:

- **Category 1** - minimal impact on the travelling public, are typically located on simple terrain, and involve two-lane highways or roads, often with lower speeds and traffic volumes.
- **Category 2** - may be located on higher-speed or higher-volume corridors and involve some complexity. Impacts on the travelling public may be moderate because of the roadway characteristics or the type of work.
- **Category 3** - complex and have a significant impact on the travelling public because of factors such as higher volumes and speeds, project duration, active night work, mountainous terrain, and/or a requirement for lane closures and/or detours.

3.3.1 Initial Project Category Assessment

The initial project category assessment considers road and traffic characteristics, as well as specific work activities.

Table 3.1: Initial Project Category Assessment on the following pages is used to determine the initial project category.

The total point value calculated at the end of Table 3.1 indicates that the project is initially assessed as a Category 1, 2, or 3.

Table 3.1: Initial Project Category Assessment

Traffic Consideration	Value	Point Value	Score
Posted or Statutory Speed Regular posted speed limit of the roadway	≤ 50 km/hr	1 point	
	60 - 70 km/hr	3 points	
	≥ 80 km/hr	4 points	
Traffic Volume Traffic volume (both directions) in peak hours	< 1,000 vehicles/hr	1 point	
	1,000 to 3,000 vehicles/hr	3 points	
	> 3,000 vehicles/hr	4 points	
Lanes Number of lanes in both directions (including auxiliary lanes)	2 lanes	0 point	
	3 lanes	2 points	
	4 lanes or more	3 points	
Encroachment Location of work	Off roadway	0 point	
	Shoulder work/partial lane closure	3 points	
	Full lane closure, ramp closure, or intersection closure	4 points	
Detours	No detour during construction	0 point	
	Detour traffic on temporary roadway during construction next to work zone.	3 points	
	Detour route during construction takes traffic off regular route away from work zone; requires detour signing	4 points	
Duration of Work	Short-duration work (no more than one day-time shift).	1 point	
	Long-duration work (less than 2 weeks)	2 points	
	Long-duration work (2 or more weeks)	4 points	
Allowable Delays Delay time plus time to travel through work zone in minutes	< 20 minutes	1 point	
	≥ 20 minutes	3 points	
	No allowable delay	4 points	

Traffic Consideration	Value	Point Value	Score
Time of Day Time of day that work will occur	Day-time only work	1 point	
	Active day-time work, with traffic control devices in place at night	3 points	
	Active night-time work	4 points	
Vertical Alignment	Flat terrain	0 point	
	Rolling terrain	1 point	
	Mountainous terrain	2 points	
Horizontal Alignment	Tangent	0 point	
	Horizontal curves, no curve advisory speeds	1 point	
	Horizontal curves, with curve advisory speeds	2 points	
Intersections	No intersections or stop-controlled intersection(s)	0 point	
	Signalized intersection(s) with no left or right turn phases, or single lane roundabout	2 points	
	Signalized intersection(s) with left or right turn phase(s), or multi-lane roundabout	4 points	
	Interchange(s)	5 points	
Runaway Lanes	No runaway lanes	0 point	
	Runaway lanes in or near the work zone; they will not be blocked at any time during course of work	1 point	
	Runaway lanes in or near work zone; they may be blocked by work or queues during course of work	4 points	
Pedestrians and Cyclists	No pedestrians or cyclists	0 point	
	Possible pedestrians and cyclists	2 points	
	Designated cycle route, sidewalk or multi-use pathway	3 points	

Traffic Consideration	Value	Point Value	Score
HOV or Bus Lane	No HOV or bus lane	0 point	
	HOV or bus lane	4 points	
Counter-Flow Lane	No counter-flow lane	0 point	
	Counter-flow lane	4 points	
		Total Score	
		Category 1	< 16
		Category 2	16 to 25
		Category 3	> 25
		Initial Project Category	

3.3.2 Project Risk Analysis

A project risk analysis is the process of reviewing site-specific characteristics and considering the likelihood and consequence of each item listed. It is able to highlight potential hazards that are not captured in the Initial Project Category Assessment.

Each project has a unique combination of site-specific characteristics, and the risk analysis considers potential hazards associated with the specific project and/or location.

Table 3.2: Project Risk Analysis on the following pages is used to determine whether each potential hazard creates a low, medium, or high risk for the project and location.

The total point value calculated at the end of Table 3.2 indicates that the project is assessed as a low-risk, medium-risk, or high-risk project.

Combining the results of the initial project category assessment and the risk analysis will determine the final project category (see *Section 3.3.3: Final Project Category Determination*).

Table 3.2: Project Risk Analysis

The Project Risk Analysis is a general guideline, applicable to most projects. If significant project-specific hazards are not included in the risk analysis below, the Evaluator may consider increasing the final risk rating. This modification and the justification for it should be documented.

All high-risk, project-specific hazards should be addressed and mitigated in the Traffic Management Plan.

Item	Risk	Definition	Point Value	Score
Falling object	Low	Potential of falling object through course of work (i.e., overhead works, slung loads, or equipment boom/bucket work)	1 point	
	Medium	Working within a known avalanche or rock fall area; no recent evidence of activity	2 points	
	High	Recent evidence of rock or material entering work site or overhead work that may impact travelling public or worker safety (i.e., overhead structures) Vehicle queues may back into a rock fall or avalanche area	3 points	
Nature of work activity	Low	Work activity is not expected to create a significant hazard	1 point	
	Medium	Work activity will create excessive dirt, dust, or gravel on the road surface, and will thereby create a potential hazard	2 points	
	High	Work activity such as blasting, scaling, or excavation < 2 metres from active travelling lanes will create a potential hazard	3 points	
Removal of safety devices	Low	No removal of safety devices	1 point	
	Medium	Removal of safety devices such as pavement markings, signage, traffic signal, or reflectors	2 points	
	High	Removal of containment devices, such as barrier, guard rail, crash attenuators, fencing, etc.	3 points	
Equipment movement through work zone	Low	Minimal conflict with traffic (e.g., work commencing off travelled roadway)	1 point	
	Medium	Conflict with normal traffic flow; no queuing or traffic stoppages	2 points	
	High	Conflicts with normal traffic; may create queuing and require traffic stoppages. Difficult for equipment to enter and exit site	3 points	

Item	Risk	Definition	Point Value	Score
Roadway surface condition during construction	Low	Roadway surface is maintained	1 point	
	Medium	Roadway surface, such as milling and grinding (consistent surface), creates a hazard for road users	2 points	
	High	Roadway surface is inconsistent, with multiple changes or work tasks (manholes, culvert installation, etc.)	3 points	
Storage of equipment and material	Low	Stored outside the shoulder	1 point	
	Medium	Stored on the shoulder but outside travelled roadway	2 points	
	High	Stored on shoulder but encroaching on travelled roadway	3 points	
Load restrictions as a result of construction	Low	No load restrictions	1 point	
	Medium	Narrow lanes restrict wide loads	2 points	
	High	Overweight/overheight vehicles restricted (may result in structural damage)	3 points	
Lane widths	Low	Maintain existing lane widths	1 point	
	Medium	n/a	n/a	
	High	Lane width not maintained throughout work zone, or Single-lane alternating traffic	3 points	
Work zone or queues block access (active or inactive site)	Low	None	1 point	
	Medium	Side street or business access	2 points	
	High	Major public facility and/or major secondary roadway	3 points	
Transit access	Low	No transit or school bus stops	1 point	
	Medium	Community shuttle or school bus stops	2 points	
	High	Express transit or major bus route	3 points	
Impacts of special events	Low	No known event	1 point	
	Medium	Moderate public event with attendance under 5,000	2 points	
	High	Major public event with attendance over 5,000 or moderate public event (under 5,000) with no alternative access or route	3 points	